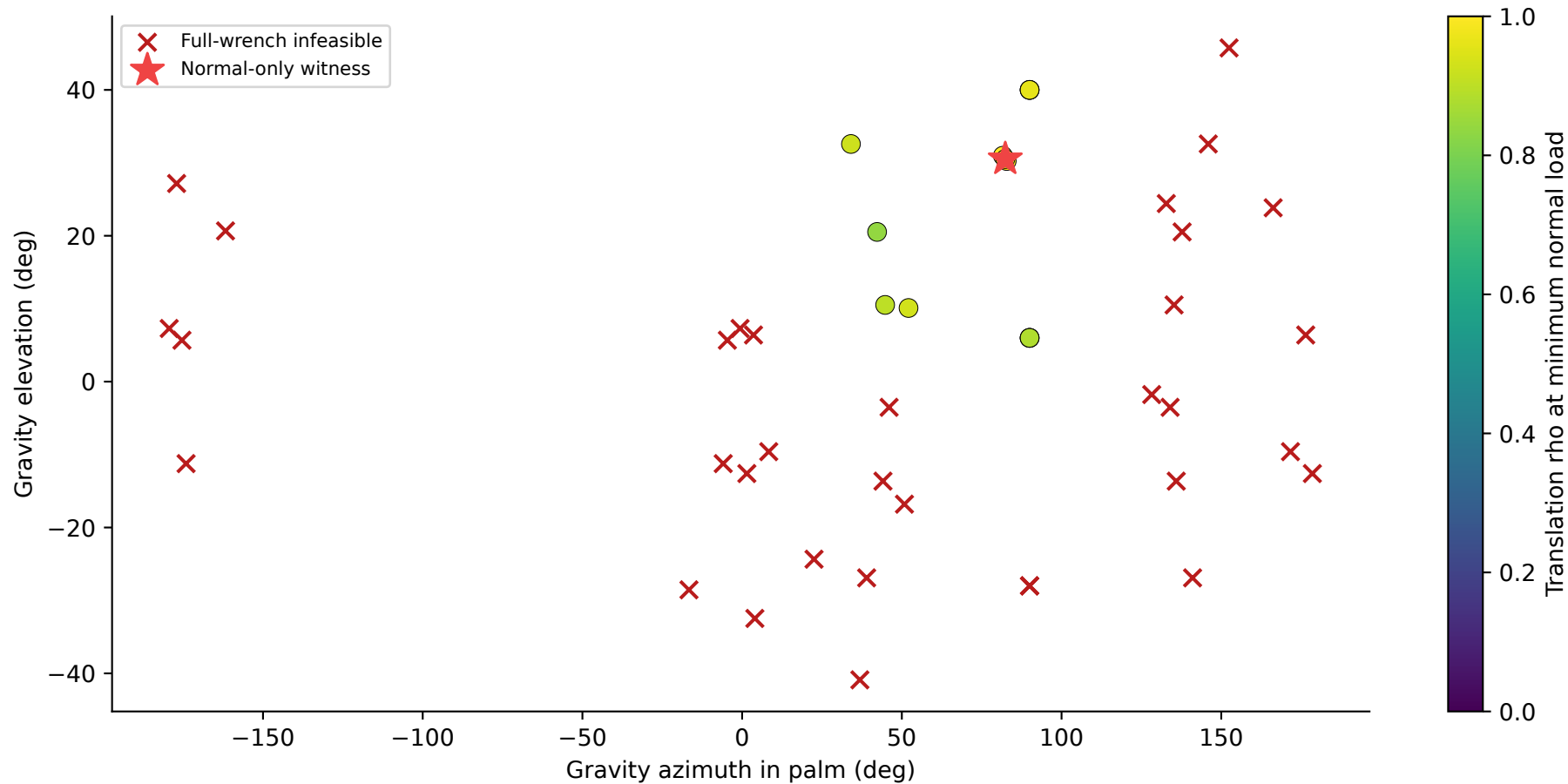


True thumb-assisted state: reachable gravity and friction



13/48 full-wrench feasible; 3/48 frictionless. Nonuniform samples, not region volume.